Teacher Wu Kuan | See It at a Glance #11 — sin 20° = cos 70° (Complementary Angles)

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1. The trap question

Is sin 20° = cos 70°?

2. The method

• The two acute angles of a right triangle add to 90°.
• Opposite side of one angle = adjacent side of the other.
• sin x = cos(90° − x), cos x = sin(90° − x). A favorite Digital SAT item.

3. Parent check method

Ask your child to label one triangle and read off sin 20° and cos 70° from the same sides.

Practice

1. sin 30° = cos ?
2. cos 25° = sin ?
3. Acute x satisfies sin x = cos 40°. Find x.
4. In right triangle ABC (C = 90°), sin A = 0.6. Find cos B.
5. Simplify sin(90° − θ).

Answers

1. 60°
2. 65°
3. 50°
4. 0.6
5. cos θ

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